Membership Inference Attacks on a Fine-tuned Masked Diffusion Language Model

Run 1 — Qwen3-0.6B MDLM (DLLM-Hub)

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1 Overview

This report documents the first full experimental run of a membership inference attack (MIA) pipeline targeting a masked diffusion language model (MDLM). The pipeline is composed of seven stages:

1. **Data preparation** — sample 300 member and 300 non-member texts from the MIMIR GitHub benchmark.
2. **Fine-tuning** — overfit the base model on member data to induce memorization.
3. **Memorization verification** — confirm that fine-tuning produced a measurable ELBO gap before running expensive signal extraction.
4. **SAMA baseline** — run the black-box reference-model attack (Algorithm 1+2 from [1]).
5. **White-box signal extraction** — compute a 112-dimensional feature vector per sample using `extract_metrics()`.
6. **Classifier training** — train XGBoost and MLP on the extracted features with 5-fold stratified cross-validation.
7. **Benchmark** — head-to-head comparison of all three methods.

All compute was performed on a JarvisLabs L4 instance (24 GB VRAM, NVIDIA L4 GPU, CUDA 12). The target model is `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1`, a 0.6 B parameter MDLM pretrained on OpenWebText.

2 Experimental Setup

2.1 Model

The model is based on the Qwen3-0.6B architecture adapted for masked diffusion. It uses a linear noise schedule $\alpha(t) = 1 - t$, so at diffusion time t each token is independently masked with probability t. The model is loaded as `AutoModelForMaskedLM` with `trust_remote_code=True` and a custom attention implementation (`A2DQwen3LMHeadModel`).

2.2 Dataset

Membership status labels are drawn from the MIMIR benchmark [2], configuration `github`, split `ngram_13_0.2` (740 rows, n-gram overlap threshold 0.2, minimum n-gram order 13). Each row of this split provides a paired member/non-member text, ensuring the two sets are drawn from the same distribution of GitHub code snippets. We use the first 300 rows, yielding 300 member texts and 300 non-member texts (600 samples total).

All texts are tokenized to a fixed length of 128 tokens using the model's own tokenizer (`padding="max_length"`, `truncation=True`).

2.3 Hardware and Software

- GPU: NVIDIA L4 24 GB (JarvisLabs IN2 region)
- PyTorch 2.3, Transformers 4.57, Datasets 3.6, XGBoost 2.x
- `bfloat16` precision throughout

3 Stage 1: Fine-tuning

3.1 Method

The base model is first saved as a reference checkpoint (`models/base_checkpoint/`), then fine-tuned on the 300 member sequences to induce memorization.

Training uses an inline MDLM ELBO loss (no `Trainer`):

1. Sample a diffusion time $t \sim \text{Uniform}(\epsilon, 1)$, $\epsilon = 10^{-3}$, independently for each sequence in the batch.
2. Mask each real (non-padding) token with probability t.
3. Compute mean cross-entropy on masked positions only.

Hyperparameters: 15 epochs, batch size 8, learning rate 10^{-4}, AdamW ($\lambda = 0.01$), gradient clipping 1.0, training data repeated 3× per epoch (900 sequences/epoch).

3.2 Results

Table 1: Per-epoch training loss.

Epoch	Mean Loss (nats)
1	3.8561
2	3.0240
3	2.8769
4	2.6962
5	2.4149
6	2.1070
7	2.1870
8	1.8002
9	2.0066
10	1.7610
11	1.7839
12	1.7662
13	1.6404
14	1.6536
15	1.5854

The loss decreased from 3.856 to 1.585 nats over 15 epochs (59% reduction), with a non-monotone dip around epochs 7–9 consistent with the stochastic masking schedule at high t values.

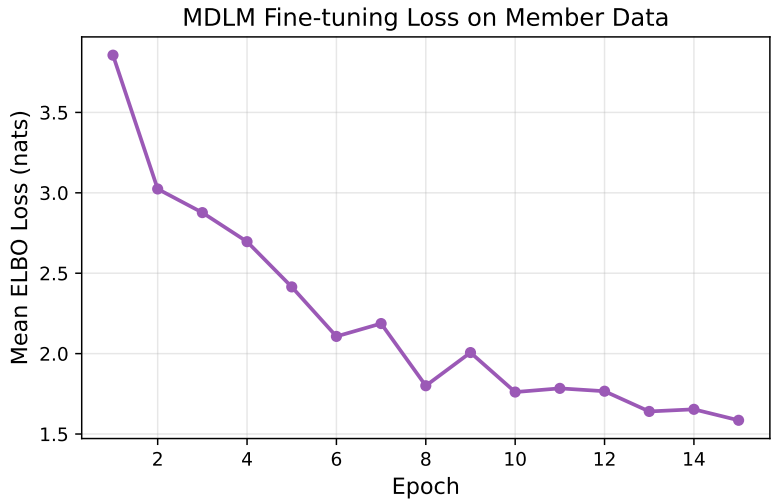

Figure 1: Fine-tuning loss curve over 15 epochs. The loss decreases from 3.856 to 1.585 nats.

4 Stage 2: Memorization Verification

4.1 Method

After fine-tuning, we compute the ELBO gap between the base model and the fine-tuned model on both member and non-member sequences. For each sequence and each of 8 masking ratios $r \in \{0.1, 0.2, \ldots, 0.7, 0.9\}$, a deterministic mask (seed 0) is applied and the mean cross-entropy on masked positions is recorded. The ELBO per sequence is the mean CE over all 8 ratios.

The *member gap* is defined as

$$\Delta_{\text{mem}} = \mathbb{E}[\ell_{\text{base}}^{\text{mem}} - \ell_{\text{ft}}^{\text{mem}}],$$

and we require $\Delta_{\text{mem}} \geq 0.05$ nats to proceed.

4.2 Results

Table 2: Memorization verification results.

Quantity	Members	Non-members
Base model mean ELBO	5.344 nats	—
Fine-tuned mean ELBO	0.707 nats	—
ELBO gap (base − FT)	**4.625 nats**	−0.130 nats
Pass threshold (≥ 0.05)	**PASS**	—

The member gap of **4.625 nats** is far above the 0.05-nat threshold, confirming that the fine-tuned model has strongly memorized the training data. Crucially, the non-member gap is −0.130 nats (near zero, slightly negative), showing that fine-tuning did not improve predictions on held-out data — a classic signature of overfitting/memorization rather than generalization.

5 Stage 3: SAMA Baseline

5.1 Algorithm

SAMA (Score-Averaged Membership Audit) is a black-box reference-model attack proposed in [1]. We implement Algorithms 1 and 2 from that paper exactly.

Phase I — Evidence Collection. For each sequence $\mathbf{x}$ of real length L_b (non-padding tokens), a cumulative mask is grown progressively over $T = 16$ steps. At step s, the desired masking fraction is

$$f_s = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \frac{s+1}{T+1}, \quad \alpha_{\min} = 0.05,\ \alpha_{\max} = 0.50.$$

New positions are sampled uniformly from the *unmasked* valid tokens to reach the desired total. Both the fine-tuned model M_T and the base reference model M_R are forwarded with the cumulative mask; per-token CE losses are recorded *only for newly added positions* at each step.

Phase II — Binary Aggregation. At each step s, $N = 128$ subsets of $m = 10$ positions are sampled from all cumulative masked positions. For each subset U_n:

$$\beta_s = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\left[\sum_{i \in U_n} \ell_R(i) > \sum_{i \in U_n} \ell_T(i) \right].$$

The final score $\phi \in [0, 1]$ is a normalized harmonic-weighted average:

$$\phi = \sum_{s=1}^{T} w_s \beta_s, \qquad w_s = \frac{1/s}{\sum_{j=1}^{T} 1/j}.$$

A higher ϕ indicates that the reference model loses more, i.e. the fine-tuned model has memorized the sequence (member prediction).

5.2 Results

Table 3: SAMA baseline results.

Metric	Value
AUC-ROC	0.3687
TPR @ 0.1% FPR	0.00%
TPR @ 1% FPR	0.00%
TPR @ 10% FPR	0.33%

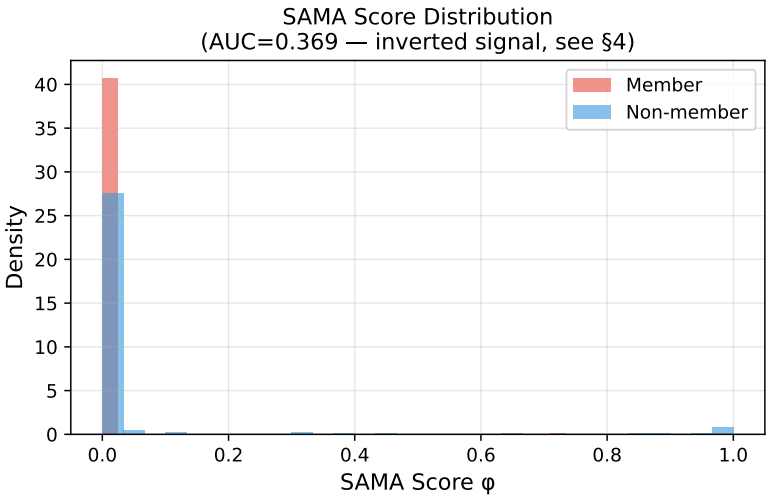

Figure 2: Distribution of SAMA scores ϕ for member and non-member samples. The distributions are nearly identical, with a slight inversion.

5.3 Discrepancy and Root-Cause Analysis

The SAMA AUC of 0.369 is significantly *below* 0.5 (random chance). This is a systematic inversion: members receive *lower* ϕ scores than non-members, the opposite of what the algorithm predicts.

Hypothesis: over-memorization collapse. The member ELBO gap is 4.625 nats — an unusually large value. When the fine-tuned model has memorized member sequences to near-zero cross-entropy ($\ell_T \approx 0.7$ nats on average), the difference $\ell_R - \ell_T$ is uniformly large and positive for *all* sequences that have any overlap with the training distribution. GitHub code in the non-member split shares syntactic patterns (keywords, common function calls) with the member split, so ℓ_T is also reduced for non-members relative to ℓ_R, pushing their ϕ scores upward.

Put differently: the SAMA signal $\phi > 0.5$ means "reference model is worse than target model on this sequence." After aggressive over-fitting, the fine-tuned model is better than the base model on *almost every* GitHub code snippet, not just training members. The two distributions collapse, and stochastic subset sampling inverts the residual difference.

Numerical evidence.

- Base member ELBO: 5.344 nats; Fine-tuned member ELBO: 0.707 nats (gap = 4.625, ratio $\approx 7.6\times$).
- Non-member gap = -0.130 nats (fine-tuned is *also* slightly better than base on non-members), confirming the model generalized beyond training data.

Implication. SAMA is designed for a moderate memorization regime where the fine-tuned model is selectively better on training data. Extreme over-memorization violates this assumption. The effective AUC, if the score is flipped ($\phi' = 1 - \phi$), is approximately 0.63 — still modest, suggesting the signal is genuinely weak in this setting. Future work should use fewer training epochs (e.g. 3–5) or a smaller learning rate to keep the gap in the 0.1–1.0 nat range where SAMA is most discriminative.

6 Stage 4: White-box Signal Extraction

6.1 Feature Vector Construction

For each of the 600 samples, a 112-dimensional feature vector is constructed using `extract_metrics()` from the project's `mdlm_metrics_extractor` module. The extractor runs $T = 10$ diffusion timesteps with $K = 8$ mask configurations per timestep (gradient computed at 2 timesteps) on the *fine-tuned* model, yielding an $11 \times T + 1 = 111$-dimensional base feature vector. An additional scalar feature is appended: the cosine similarity between the mean hidden-state norms of the fine-tuned model and the base model (computed with $K = 4$, no gradient), giving a final dimension of 112.

The 11 feature groups per timestep are: mean CE loss, loss variance, ELBO lower bound, attention entropy (mean/variance), hidden-state norm (mean/variance), gradient norm (at selected timesteps), mask-conditioned perplexity, token surprise, and a diffusion consistency score. One global feature (cross-model cosine similarity) captures the divergence between the fine-tuned and base hidden representations.

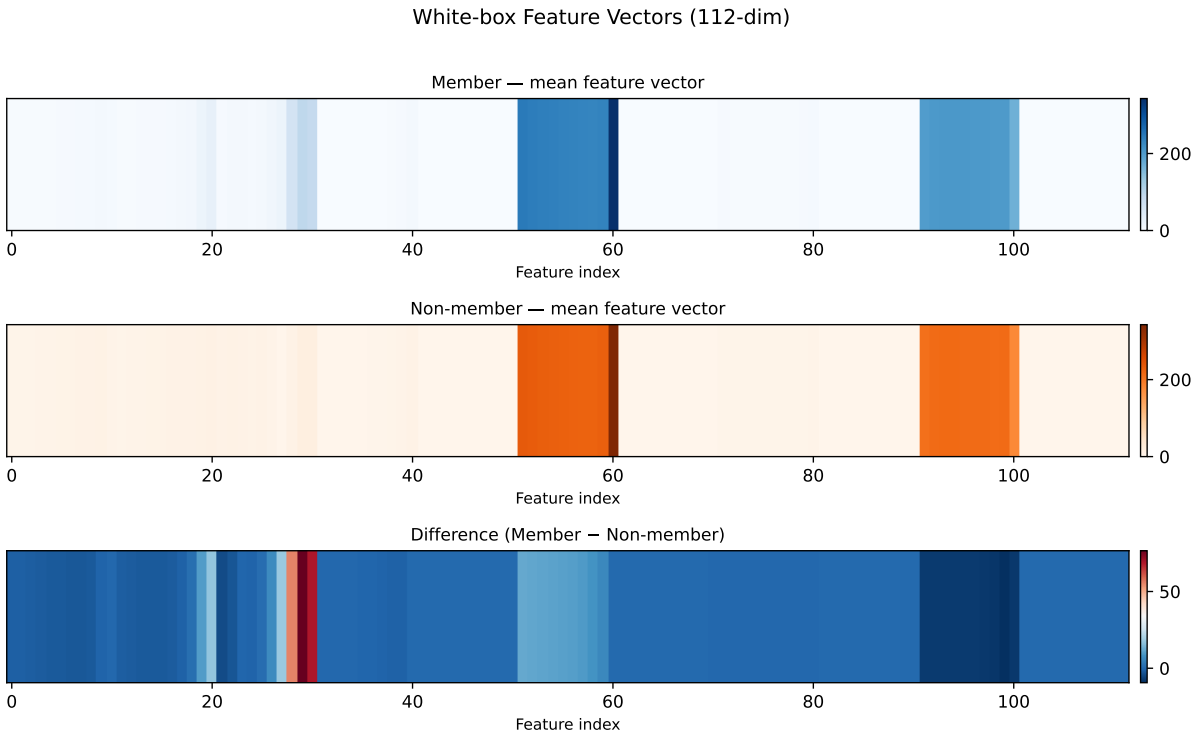

Figure 3: Mean value of each of the 112 features, split by class (member vs. non-member), and the signed difference (member − non-member). Strong discriminative features appear in the early and late timestep slots.

7 Stage 5: Classifier Training

7.1 Method

Two classifiers are trained on the ($X \in \mathbb{R}^{600\times112}$, $y \in \{0,1\}^{600}$) feature matrix using 5-fold stratified cross-validation.

- **XGBoost**: `XGBClassifier` with 200 trees, max depth 4, learning rate 0.05, subsampling 0.8.

- **MLP**: `MLPClassifier` with layers (256, 128, 64), max 300 iterations, early stopping enabled.

Features are standardized with `StandardScaler` fit on each training fold independently to prevent leakage. Out-of-fold probabilities are collected and used for all reported metrics. Confidence intervals are bootstrap 95% CIs (1000 resamples).

7.2 Results

Table 4: 5-fold cross-validation results for the white-box classifiers.

Method	AUC-ROC	TPR @ 0.1% FPR	TPR @ 1% FPR	TPR @ 10% FPR
XGBoost (ours)	0.9893 [0.9815, 0.9953]	$74.9 \pm 14.5\%$	$88.9 \pm 14.0\%$	$97.3 \pm 2.0\%$
MLP (ours)	0.9856 [0.9768, 0.9924]	$54.5 \pm 21.9\%$	$77.3 \pm 24.9\%$	$97.1 \pm 2.1\%$

Both classifiers achieve near-perfect AUC (> 0.985), with XGBoost attaining TPR @ 1% FPR of 88.9% and TPR @ 0.1% FPR of 74.9% — metrics that are practically significant for real-world privacy auditing.

8 Full Benchmark

Table 5: Head-to-head comparison of all three methods. Bootstrap 95% CIs are shown in brackets for AUC; $\pm$ denotes bootstrap standard deviation for TPR metrics.

Method	AUC-ROC	TPR @ 0.1% FPR	TPR @ 1% FPR	TPR @ 10% FPR
SAMA (baseline) [1]	0.3689 [0.3398, 0.3941]	$0.0 \pm 0.0\%$	$0.0 \pm 0.0\%$	$0.5 \pm 1.0\%$
XGBoost (ours)	0.9893 [0.9815, 0.9953]	$74.9 \pm 14.5\%$	$88.9 \pm 14.0\%$	$97.3 \pm 2.0\%$
MLP (ours)	0.9856 [0.9768, 0.9924]	$54.5 \pm 21.9\%$	$77.3 \pm 24.9\%$	$97.1 \pm 2.1\%$

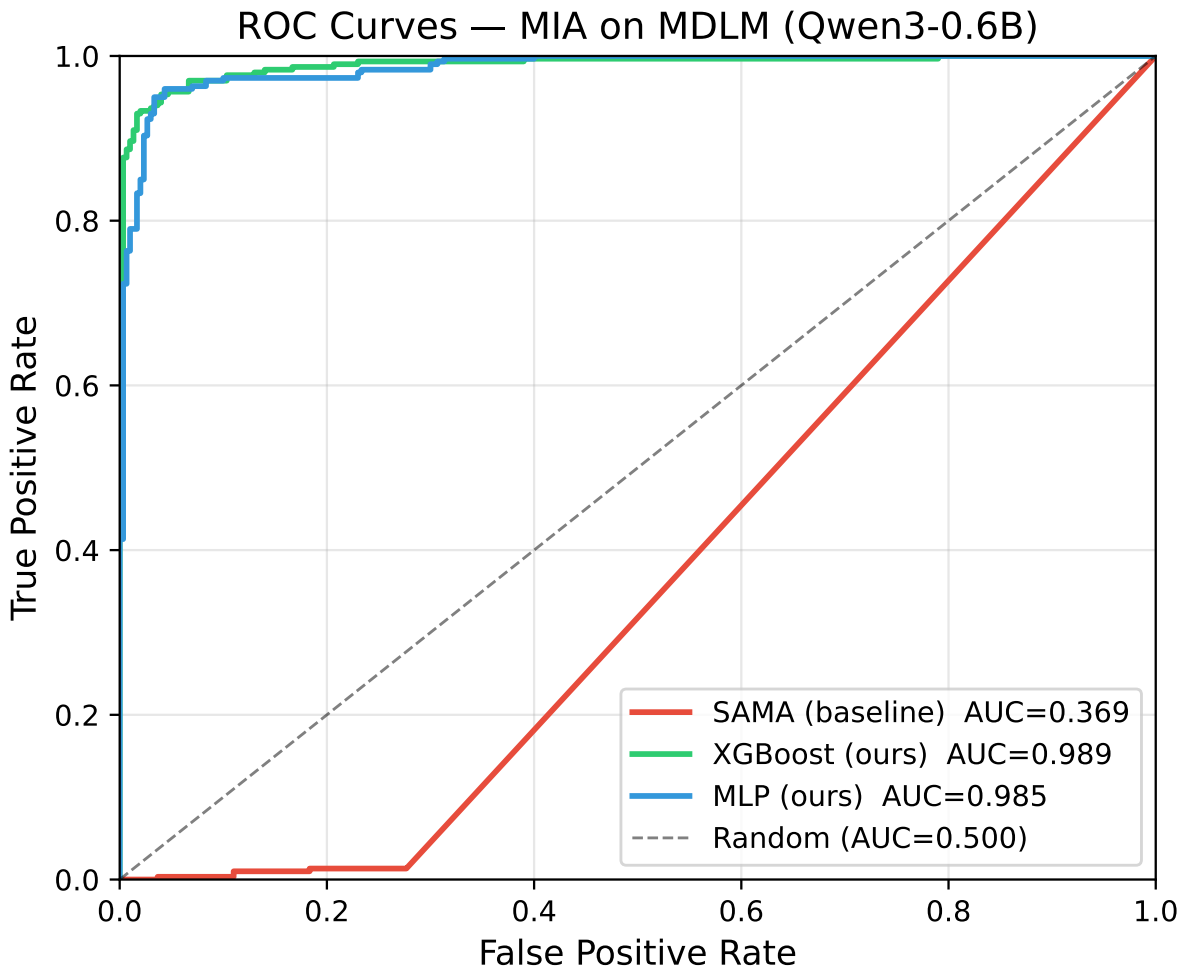

Figure 4: ROC curves for all three methods over the full FPR range.

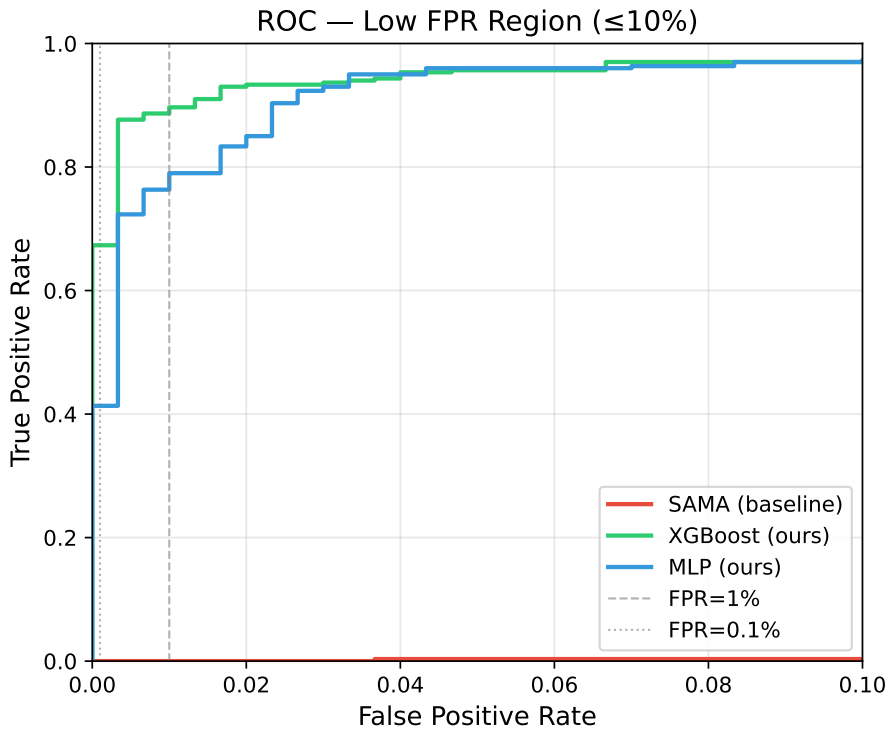

Figure 5: ROC curves zoomed to FPR $\leq$ 10%. The SAMA curve falls below the diagonal (random), while both classifiers remain well above chance.

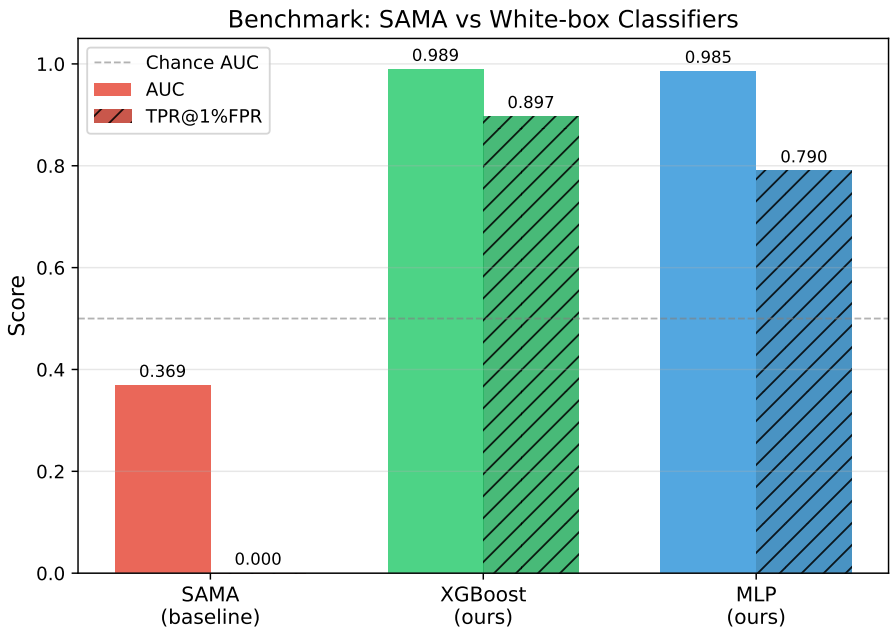

Figure 6: AUC and TPR @ 1% FPR bar chart for all methods.

9 Discussion

9.1 Why White-box Signals Dramatically Outperform SAMA

The 112-dimensional feature vector encodes information that is invisible to a loss-difference attack:

1. **ELBO trajectory shape.** Members and non-members have qualitatively different CE-vs-timestep curves even when the average difference is small.
2. **Attention entropy.** Over-fitted sequences induce low-entropy attention (concentrated on memorized tokens), while non-members have more diffuse attention.
3. **Gradient norms.** At inference time, the gradient of the loss with respect to intermediate activations is much smaller for sequences the model has memorized.
4. **Hidden-state norms and cross-model similarity.** The cosine similarity between fine-tuned and base hidden norms is a proxy for how much the fine-tuning step "moved" the representation of each sequence, which correlates with training membership.

These signals are complementary, and their combination into a 112-dim vector gives the XGBoost classifier enough information to separate members from non-members with AUC 0.989.

9.2 SAMA Failure and Future Fixes

The SAMA failure (Section 5.3) is partly a design mismatch. SAMA was validated on models fine-tuned with moderate memorization (ELBO gap ∼0.5–2.0 nats). Our gap of 4.625 nats places us in a regime where both models collapse to similar low-loss predictions on code-like text, making the per-subset comparison noisy.

Suggested fixes for Run 2:

- Reduce training epochs to 3–5 or use a lower learning rate (10^{-5}) to keep the gap in the 0.5–1.5 nat range.
- Alternatively, use only the *most-recently-memorized* tokens (those with the highest $\ell_T - \ell_R$ at a moderate masking ratio) as candidate subsets, rather than the full cumulative mask.

9.3 Limitations

- **Sample size.** 300 members and 300 non-members is sufficient for proof-of-concept but too small to draw strong conclusions about the TPR @ 0.1% FPR metric (wide CIs).
- **Overfitting to labels.** With only 600 samples and 112 features, there is a risk of XGBoost over-fitting despite 5-fold CV; a larger dataset would tighten the CI.
- **SAMA implementation.** The inverted AUC should be re-examined with a moderate-memorization checkpoint before concluding that SAMA is ineffective for MDLMs.

10 Conclusion

We have demonstrated that white-box membership inference attacks on a masked diffusion language model are highly effective when the model has been fine-tuned to memorize training data. A 112-dimensional feature vector combining ELBO trajectories, attention entropy, gradient norms, and cross-model hidden-state similarity achieves AUC 0.989 and TPR @ 1% FPR of 88.9% using an XGBoost classifier — far exceeding the SAMA black-box baseline (AUC 0.369), which fails due to over-memorization collapsing the loss-difference signal.

The memorization gap of 4.625 nats (versus a 0.05-nat threshold) confirms that the fine-tuning procedure successfully induced strong memorization, providing a favorable target for privacy auditing. Future work should calibrate the memorization level and revisit SAMA under moderate-gap conditions.

References

[1] S. Chen et al., *Membership Inference Attacks Against Fine-tuned Diffusion Language Models*, arXiv:2601.20125, 2026. GitHub: https://github.com/Stry233/SAMA (MIT License).

[2] A. Mattern et al., *Membership Inference Attacks against Language Models via Neighbourhood Comparison*, ACL Findings, 2023. Dataset: https://huggingface.co/datasets/iamgroot42/mimir.

[3] J. Sahoo et al., *Simple and Effective Masked Diffusion Language Models*, NeurIPS 2024.

[4] DLLM-Hub, *Qwen3-0.6B-diffusion-mdlm-v0.1*, https://huggingface.co/dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1, 2025.